

VISHAL BAKSHI

FL, USA — +1 (571) 579-0298 — vishal.bakshi@gwu.edu — [LinkedIn](#) — [Portfolio](#) — [GitHub](#)

Summary

Data Scientist and Machine Learning Engineer with 3+ years of experience specializing in the development and optimization of scalable ML models for real-time systems. Proven expertise in building complex data pipelines (Python, Spark) and applying deep learning (PyTorch) and statistical methods to enhance critical business signals. Experienced in time-series forecasting and translating model outputs into actionable insights for cross-functional teams to drive operational efficiency and data-driven decision-making.

Experience

Data Engineer Intern

June 2025 – Present

FCCI Insurance Group

Sarasota, FL

- Maintained core ETL pipelines for critical insurance data, resolving 10+ production bugs to reduce pipeline failures by 95% and ensure on-time delivery of daily financial reports.
- Developed a data validation report using complex SQL joins to proactively identify and flag duplicate claims, preventing an estimated \$20,000 in erroneous payouts per quarter.
- Delivered \$36,000 in business savings by building and deploying a production-grade, multi-threaded Python pipeline to automate legacy code audits using the OpenAI GPT API, reducing manual review effort by 80%.

Machine Learning Engineer

May 2024 – May 2025

HIVE Research Lab - George Washington University

Washington, DC

- Uncovered that 62.9% of healthy patients exhibited early pre-diabetic patterns by developing an ETL and data workflow for 60 GB of time-series patient data; a critical finding validated with clinical SMEs for early intervention.
- Designed and implemented a model evaluation framework focused on recall for at-risk patient identification; improved performance by 20% through engineering 5+ novel temporal features from complex CGM data.
- Proposed and designed the architecture for a real-time patient feedback system, specifying the **RESTful API endpoints** and **data schemas** required to reliably translate analytical outputs into actionable health alerts via FastAPI.

Machine Learning - Robotics Engineer (Computer Vision)

June 2022 – August 2023

HBots Robotics Pvt Ltd

Hyderabad, Telangana

- Owned the end-to-end development of a novel computer vision system, from dataset acquisition and model prototyping in PyTorch to production deployment, directly addressing a core business need for automation.
- Architected and built a custom image dataset from the ground up, implementing targeted data augmentation strategies that increased object detection model mAP by 35% and met a critical 3-month product deadline.
- Engineered a sophisticated data fusion pipeline integrating stereo depth cameras with the 2D detection model, enabling precise 3D geometric volume estimation—a key feature for the product's value proposition.

Projects

Intelliwiz - Full-Stack AI Application for Automated Data Intelligence

[Link](#)

Technologies: Python, LangChain, LangGraph, Streamlit

- Developed a production-ready agentic system using LangGraph to automate complex data intelligence workflows, proving the viability of autonomous agents for data processing tasks and reducing manual workload by over 70%.
- Architected an LLM-powered engine to auto-generate 12+ chart types (Plotly-based) from schema analysis, providing immediate visual insights without requiring any user scripting.
- Engineered and deployed the solution as an intuitive Streamlit application, enabling non-technical business stakeholders to generate executive-ready reports in minutes, bridging the gap between raw data and insights.

New York Airbnb Review Analysis

[Link](#)

Technologies: Python, BERT, T5, KMeans, Scikit-learn, AWS EC2, NLTK, spaCy

- Engineered an advanced NLP pipeline to analyze 17,000+ Airbnb guest reviews, fine-tuning BERT models for sentiment classification, achieving high contextual accuracy to enhance guest experience insights.
- Developed an Aspect-Based Sentiment Analysis system, leveraging KMeans clustering and aspect tagging to extract granular insights on cleanliness, location, and host interactions, driving data-informed decisions for hosts.
- Designed a transformer-based T5 review summarization model, automating concise, high-quality summaries of guest feedback, reducing manual review time by 60% while improving host responsiveness and service optimization.

Education

The George Washington University, Washington, DC

May 2025

Master of Science in Data Science

GPA: 3.85/4.0

Technical Skills

Languages: Python, SQL (Spark SQL, Presto), Go, Java

ML & Statistical Modeling: PyTorch, TensorFlow, Scikit-learn, Deep Learning, Time-Series Forecasting, Optimization, Probabilistic Modeling

Data & MLOps Tools: Apache Spark, AWS (SageMaker), GCP, Docker, Kubernetes, MLflow, CI/CD (GitHub Actions)

Libraries & Development: LangChain, Hugging Face, Git, Agile